USER GUIDE / 01

CLARITAS

USER GUIDE

Windows with VS Code
Linux setup and operation

A practical guide to the cleaned single-file CUDA Monte Carlo model. Includes environment setup, build and run commands, first-case configuration, sweeps, output interpretation and troubleshooting.

Start here	Section
Existing Windows environment	2 - Windows / VS Code
New Windows environment	3 - Windows installation
Linux environment	4 - Linux installation
Build and run	5 - Validation and execution
Small first case	6 - First production run
Configuration and science	7-10
Diagnostics and results	11-13
Problems, reference and all defaults	14 onward

Already configured?

Open the extracted project folder in VS Code, activate your existing **claritas** environment, save any configuration edits, and run these commands in its terminal. No installation is needed if the environment is working.

```
python -m py_compile CLARITAS.py
python validate_claritas.py --cuda-full
python test_diagnostics.py
python -u CLARITAS.py
```

There is no separate executable build. CuPy compiles the embedded CUDA kernels when the model is instantiated; the CUDA tests exercise that route. Syntax compilation alone does not build or validate CUDA.

Guide date: 11 October 2026. Based on the delivered CLARITAS.py, requirements.txt and current validation scripts. No development chronology is needed to use this implementation.

USER GUIDE / 02

Windows: VS Code and an existing environment

1. Open the right folder

Extract the publication package into a project folder, for example **C:\Projects\CLARITAS**. The folder must directly contain CLARITAS.py, requirements.txt, validate_claritas.py and test_diagnostics.py. In VS Code choose **File > Open Folder** and open that folder.

2. Select the Python interpreter

Install the Microsoft Python extension if it is absent. Press **Ctrl+Shift+P**, choose **Python: Select Interpreter**, and select the Python executable in your **claritas** Conda environment. Interpreter selection and terminal activation are separate checks. Open a **new** terminal after selection. [2]

3. PowerShell terminal commands

```
Set-Location "C:\Projects\CLARITAS"
conda activate claritas
python -c "import sys; print(sys.executable)"
python --version
nvidia-smi
python -m pip check
```

The executable path should identify the selected environment. If it points to a different Python installation, activate/select the intended environment before proceeding. The example folder is replaceable; keep quotes around paths containing spaces.

4. Build, test and run

```
python -m py_compile CLARITAS.py
python validate_claritas.py --cuda-full
python test_diagnostics.py
python -u CLARITAS.py
```

Run one command at a time and inspect its result. A CUDA SKIP means the GPU tests did not execute. Avoid the Code Runner extension for this workflow: its interpreter or working directory may differ from the activated terminal.

Git Bash alternative

In VS Code choose **Terminal: Select Default Profile > Git Bash**, then open a new terminal. A Windows path such as C:\Projects\CLARITAS is written as /c/Projects/CLARITAS. The build/test/run commands above are unchanged.

```
cd "/c/Projects/CLARITAS"
conda activate claritas
python -c "import sys; print(sys.executable)"
```

Use the shell that already works with your Conda environment. If Conda activation is unavailable, follow the shell-initialisation step on the next page.

USER GUIDE / 03

Windows: first-time environment setup

Skip this section when your existing environment is working. The commands below create a new isolated environment; they are not a request to reinstall your current setup.

Install the prerequisites

• A compatible NVIDIA GPU and installed NVIDIA driver. Confirm that **nvidia-smi** runs. A driver is required even when CUDA components are supplied by Python packages.

• VS Code for Windows and its Microsoft Python extension. The Windows user installer is a suitable route. [3]

• Miniforge (Conda) for Windows x86-64, or your existing Miniconda installation. Use the official installer and open its installed Conda prompt. [4]

Enable Conda in the intended VS Code shell

From the installed Conda prompt, initialise PowerShell, then close and reopen VS Code and the terminal. Conda initialisation changes the shell profile. [5]

```
conda init powershell
```

For Git Bash, use **conda init bash** instead. If your organisation restricts PowerShell profiles, use an approved Conda-capable terminal or Git Bash; do not bypass its policy.

Create and populate the environment

```
conda create -n claritas python=3.12 -y
conda activate claritas
Set-Location "C:\Projects\CLARITAS"
python -m pip install -r requirements.txt
python -m pip check
```

If an environment named claritas already exists, activate it rather than recreating it. Select that interpreter in VS Code as described in section 2.

What the supplied requirements install

Dependency	Supplied requirement
NumPy	numpy>=1.26,<3
Matplotlib	matplotlib>=3.8,<4
CuPy with CUDA 12 components	cupy-cuda12x[ctk]>=14,<15

The [ctk] extra supplies CUDA components with the CuPy wheel. A separate system-wide CUDA Toolkit is not required for this wheel-based route when it resolves and runs successfully; the NVIDIA driver still must be compatible. Do not install multiple CuPy distributions into the same environment. [1]

The CUDA version displayed by nvidia-smi describes driver capability, not the installed Python package. Do not replace the supplied CUDA-12 wheel solely because nvidia-smi reports a newer CUDA capability.

USER GUIDE / 04

Linux: native setup and VS Code

These instructions target a native Linux x86-64 system with an NVIDIA GPU. Linux and Windows environments are separate; do not reuse a Windows Python environment from Linux or WSL.

1. Check the GPU and prepare the project

```
nvidia-smi
mkdir -p "$HOME/projects/CLARITAS"
cd "$HOME/projects/CLARITAS"
```

Extract or copy the publication package into this folder. For an archive in your Downloads directory, Python can extract it without requiring an unzip utility:

```
python3 -m zipfile -e   "$HOME/Downloads/CLARITAS_Publication_Ready.zip" .
```

If nvidia-smi fails, resolve the driver/device issue using your distribution and NVIDIA guidance [7]. Python installation does not supply the NVIDIA driver.

2. Create the environment

If Conda is not installed, obtain the official Linux x86-64 Miniforge installer [4], run it with bash, allow shell initialisation and reopen your terminal. If Conda already works, skip installation.

```
bash "$HOME/Downloads/Miniforge3-Linux-x86_64.sh"
# Run the installer only if Miniforge is not installed.
```

```
conda create -n claritas python=3.12 -y
conda activate claritas
cd "$HOME/projects/CLARITAS"
python -m pip install -r requirements.txt
python -m pip check
python -c "import sys; print(sys.executable)"
```

3. Open in VS Code

Install VS Code from its official Linux package instructions [6] and add the Microsoft Python extension. Open the project with **code .**, or use File > Open Folder. Select the claritas interpreter and open a new terminal. Run the commands in section 5.

Alternative: an existing Python 3.12 plus venv

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python -m pip check
```

Use either Conda or venv, not both for one run. This venv route assumes Python 3.12 and its venv support are already installed. For headless plotting, set **export MPLBACKEND=Agg** and leave SHOW_PLOTS=False.

USER GUIDE / 05

Build, validate and execute

Check Python, GPU visibility and allocation

```
python -c "import sys; print(sys.executable)"
python -c "import cupy as cp; cp.show_config()"
python -c "import cupy as cp; print(cp.cuda.runtime.getDeviceCount())"
python -c "import cupy as cp; cp.cuda.Device(0).use(); print(cp.arange(5).get())"
```

The allocation command tests device 0 and should print [0 1 2 3 4]. If your configured GPU_DEVICE is different, substitute that index. This is an environment check, not a scientific transport result.

Command	Purpose / expected scope
python -m py_compile CLARITAS.py	Python syntax check; no CUDA compilation.
python validate_claritas.py	16 host/static checks; four GPU checks skipped.
python test_diagnostics.py	10 CPU tests for policies, accounting, exports and mocked sweeps.
python validate_claritas.py --cuda	CUDA compilation/parity and diagnostic kernels; solver/full-ray checks skipped.
python validate_claritas.py --cuda-full	Adds stationary solver and small end-to-end GPU ray checks.
python -u CLARITAS.py	Runs the saved configuration; -u gives unbuffered console output.

Capture a run log

This redirection works in PowerShell, Git Bash and Linux Bash. It writes a log instead of displaying continuous progress; use a distinct log name for each run.

```
python -u CLARITAS.py > claritas_run.log 2>&1
```

PowerShell: inspect with **Get-Content claritas_run.log -Tail 40 -Wait** in another terminal. Linux/Git Bash: use **tail -f claritas_run.log**. The monitoring command can be stopped independently.

Stop and restart behavior

Ctrl+C requests interruption. Previously completed case folders and checkpoints can remain, but an interrupted case may be incomplete. There is no automatic resume from sweep_progress.json. Rerunning starts cases again and uses new session/case directories rather than silently overwriting existing results.

During preparation of this guide, host tests were available; CUDA was unavailable. A device PASS must come from your GPU, not from the guide or a skipped test.

USER GUIDE / 06

A small first production case

Use a short case to check execution, output paths and plotting before committing to the supplied six-case production workload. This example reduces sampling and diagnostic storage; it does not change the configured physical model.

Edit and save these settings in CLARITAS.py

```
RUN_SWEEP = False
MATERIAL = "loess"
CONCENTRATION_G_L = 0.5
WAVELENGTH_NM = 622.0
N_RAYS = 100_000
LIGHT_TRANSPORT_DIAGNOSTIC_MAX_RAYS = N_RAYS
LIGHT_TRANSPORT_GRID_NX = 64
LIGHT_TRANSPORT_GRID_NY = 64
LIGHT_TRANSPORT_GRID_NZ = 96
LIGHT_TRANSPORT_REPRESENTATIVE_RAYS = 64
LIGHT_TRANSPORT_MAX_SEGMENTS_PER_RAY = 128
OUTPUT_DIR = "claritas_smoke_output"
SHOW_PLOTS = False
```

Replace the corresponding assignments; do not append a second configuration block after main(). Keep the geometry, optical properties, vortex and sediment model unchanged for this first check.

Validate configuration without starting CUDA

```
python -c "import CLARITAS as c; c._validate_user_configuration(); print('Config OK')"
python -m py_compile CLARITAS.py
python -u CLARITAS.py
```

Check the saved products

• Confirm the console identifies your GPU, 100,000 rays, 98 mm bore and the intended model.

• Find run_config.json and run_summary.json in the case folder. Confirm requested_n_rays, seed, geometry and model_kwargs.

• Read production_integrity_report.json and light_transport_integrity_report.json. A finished run can still have a limited diagnostic status.

• Check detector_response.csv and the saved figures. Sparse counts or missing rear-angle detections in a small case are not evidence of convergence.

Return to the supplied workload

Restore RUN_SWEEP=True, N_RAYS=100_000_000, the 256 × 256 × 384 diagnostic grid, 256 representative rays, 512 segments and OUTPUT_DIR="claritas_output". Restore LIGHT_TRANSPORT_DIAGNOSTIC_MAX_RAYS=N_RAYS. Save before running again.

Changing N_RAYS after importing the module does not automatically update a value already assigned from it. For normal use, edit the source and start a new Python process.

USER GUIDE / 07

Single cases, sweeps and reproducibility

Select the run mode

RUN_SWEEP=False: MATERIAL, CONCENTRATION_G_L and WAVELENGTH_NM define one case.
RUN_SWEEP=True: MATERIALS, CONCENTRATIONS_G_L and WAVELENGTHS_NM define a Cartesian product in that order. The single-case settings are then not used to select cases.

```
RUN_SWEEP = True
MATERIALS = ["loess", "kaolin"]
CONCENTRATIONS_G_L = [0.5, 2.0, 4.0]
WAVELENGTHS_NM = [622.0]
RANDOM_SEED = 24620001
USE_SAME_SEED_FOR_SWEEP = False
```

Case	Material / concentration	Seed
1	loess / 0.5 g/L	24620001
2	loess / 2 g/L	24620002
3	loess / 4 g/L	24620003
4	kaolin / 0.5 g/L	24620004
5	kaolin / 2 g/L	24620005
6	kaolin / 4 g/L	24620006

With USE_SAME_SEED_FOR_SWEEP=False, the seed is RANDOM_SEED plus the zero-based case index. Reordering or adding cases changes those assignments. True uses the same seed for all cases; it does not guarantee identical histories across changed physical inputs.

Reproduce a case

Record the source file, run_config.json, source hash, dependency versions, GPU, driver, ray count, seed and configuration. Preserve CHUNK_SIZE and STATISTICS_BATCH_RAYS when comparing executions. GPU atomic reductions can change low-order floating-point results despite identical seeds.

```
python -m pip freeze > environment_packages.txt
python -c "import CLARITAS as c; print(c._current_source_sha256())"
nvidia-smi
```

Detailed sediment export in a sweep

DETAILED_DIAGNOSTICS_PER_SWEEP_CASE=False requests detailed sediment products for the first completed case of each material; later cases still produce the regular case outputs and summary products. Set True when you need detailed fields for every case. Single-case runs are detailed.

Convergence is a separate task

Use independent seeds and increase ray count to assess Monte Carlo uncertainty, particularly for weak detector channels. Study stationary-grid resolution and solver residual separately. A small execution check or a matching seed does not establish physical accuracy.

USER GUIDE / 08

Geometry, units and current settings

Quantity	Current setting / meaning
Internal water bore	Diameter 0.098 m; radius 0.049 m.
Wall and water	Cell height 0.491 m; water volume-mean level 0.142 m above internal bottom.
Sensor plane	0.093 m above bottom; internal CUDA z=0 is this plane.
Internal axial coordinates	Bottom z=-0.093 m; volume-mean water level z=0.049 m; wall top z=0.398 m.
Acrylic wall	Thickness 0.0035 m; outer radius 0.0525 m.
Ring radii	Inner radius 0.0530 m; outer radius 0.0680 m from wall/clearance/width.
Source launch	SOURCE_LAUNCH_RADIUS_M=None follows ring outer radius (0.0680 m).
Source direction	Launch is at negative y, facing +y, at the sensor plane; Gaussian spot and selected angular law.
Detector channels	0 to 170 degrees in 10-degree steps; finite hardware aperture tracing.

Interpret concentration and size

Concentration is in g/L and is the total sediment mass loading integrated over the actual water volume. Spatial sediment transport redistributes that loading by size; local sensor concentration can differ from the bulk input. Particle arrays store **diameters in metres**, not radii. The supplied source has 38 loess bins and 37 kaolin bins.

Free surface and connected acrylic

WATER_LEVEL_M controls volume-mean height. A vortex changes centre and wall heights while preserving the selected surface volume. The acrylic sidewall and bottom disc form a connected solid; the water aperture at the bottom disc is an interface only inside the bore. Free-surface transmission can enter the tube headspace before an external-air exit.

Current versus supplied comparison

The comparison dictionaries use a 93 mm bore, finite-Re drift/diffusion and zero meridional circulation. Current defaults use a 98 mm bore and meridional advection/diffusion with circulation ratio 0.5. False comparison-match flags are expected for those differences and are not an execution failure.

Changing bore size propagates wall and ring radii through the geometry resolver. Changing heights must keep the detector plane submerged and the prescribed free surface inside the finite vessel. Configuration validation checks those relationships.

USER GUIDE / 09

Stationary sediment and vortex controls

Setting / option	Physical interpretation
`SEDIMENT_DISTRIBUTION="uniform"`	Homogeneous suspension; vortex optical geometry may remain enabled.
`"steady_state"`	PSD-resolved stationary field with transport balance; no time marching.
`stokes_drift_diffusion`	Stokes slip response for settling and vortex-driven radial drift.
`schiller_naumann_drift_diffusion`	Finite-Re drag correction with analytic/lookup drift-diffusion field.
`schiller_naumann_meridional_advection_diffusion`	Finite-Re slip plus steady incompressible bulk circulation; CUDA finite-volume solver.
`localized_scully_vortex`	Localized core with VORTEX_CORE_RADIUS_M; prescribed surface and matching drift.
`parabolic_vortex`	Rigid-body rotation with a volume-preserving parabolic free surface.

Main physical parameters

SEDIMENT_EDDY_DIFFUSIVITY_M2_S is the diffusion coefficient (default 0.001 m^2/s). Particle density is 2600 kg/m^3, water density 997 kg/m^3, viscosity 8.9e-4 Pa s and gravity 9.81 m/s^2. MERIDIONAL_CIRCULATION_RATIO scales the circulation speed relative to the selected vortex peak azimuthal speed; its allowed range is 0 to 0.5.

The meridional model requires an enabled, nonzero supported vortex. A positive circulation ratio gives outward flow near the bottom, upward flow in the outer annulus, inward return near the free surface and downward axial return.

Resolution and solver controls

MERIDIONAL_GRID_NR and MERIDIONAL_GRID_NZ are production sediment-grid controls, distinct from the light-history grid and reporting grids. MAX_ITERATIONS, CHECK_INTERVAL, TOLERANCE and SOR govern stationary convergence. Failure to converge is reported rather than silently accepting a field.

SEDIMENT_TRANSPORT_QUADRATURE_POINTS and SEDIMENT_RADIAL_LOOKUP_POINTS control startup integration and radial lookup. SEDIMENT_OUTPUT_NR/NZ and profile controls sample results for reporting; they do not replace the production sediment discretisation.

What steady state means

The field satisfies a stationary balance of settling, diffusion, radial drift and optional meridional advection. It does not predict how quickly sediment settles, how a vortex forms, or how concentration changes after a transient disturbance. Adding a time series of stationary runs does not create a transient solver.

USER GUIDE / 10

Optics, phase functions and advanced options

Keep the three channels distinct

• **Geometric:** a finite rough particle interface with sampled placement, Fresnel interactions and internal transport.

• **Coherent Mie:** direction sampling from the selected PSD bin's Mie phase table.

• **Morphology:** an incoherent rough dielectric angular event at the macroscopic scattering site; it does not translate the ray by a physical sphere chord.

The configured model template selects hybrid_mie_exact_rate with unit wave scale, 25-degree RMS surface slope, 100 nm wave RMS height, and isotropic spherical morphology. The source radiance law is beta_radiance_3d; absorbing bores and ideal_hard_aperture photodiode response are selected.

Material indices and wavelength

PARTICLE_N_REAL_OVERRIDE=None selects the stored size-dependent n(d) profile. PARTICLE_K_IMAG_OVERRIDE=0.0 explicitly sets nonabsorbing particles; it does not select the stored exploratory k(d) arrays. Overrides may be scalars or arrays matching the selected material bin count. One common array cannot generally cover both 38-bin loess and 37-bin kaolin sweeps.

Mie efficiencies, phase tables, coherence and internal absorption use wavelength explicitly. The model has no automatic spectral dispersion for water, acrylic or particle n(d). An arbitrary wavelength sweep therefore retains the configured indices unless you supply a justified optical basis.

Reporting phase function

phase_function.csv contains an event-weighted Mie CDF and its numerical angular derivative, labelled per radian. It is an angular reporting average, not a complete hybrid phase law and not the per-bin production sampler. Do not substitute it for the bin-specific CDFs or assume it is a per-steradian density.

Programmatic options

ClaritasForwardModel and ReportingClaritasForwardModel expose event laws, source alternatives, aggregation/fragmentation, spheroid orientation, detector models and optional moment constraints. These are constructor arguments; the script does not accept command-line model flags. MODEL_DEFAULT_KWARGS supplies the template and _build_model applies current user controls.

legacy_polar_pdf is an active polar-PDF comparison source, and legacy_* detector fields describe a separate angular-window estimator. Moment-constrained runs require valid per-material runtime coefficients; selecting that option alone is not a calibrated model. Advanced changes require their own scientific validation.

USER GUIDE / 11

Light-history diagnostics and resource controls

What is collected

The diagnostic path observes actual production rays without consuming RNG or altering transport decisions. Exact voxel traversal deposits in-water track length, post-scatter length and particle-event fields. A bounded subset also stores representative trajectory segments.

Control group	Effect
`N_RAYS`	Number of launched production rays, per case.
`CHUNK_SIZE`	GPU ray-buffer chunk size; reduce to manage transport-buffer memory.
`STATISTICS_BATCH_RAYS`	Batch grouping for statistics and jackknife estimates; not a spatial resolution.
`LIGHT_TRANSPORT_DIAGNOSTIC_MAX_RAYS`	Bound on observed histories; defaults to N_RAYS at module load.
`LIGHT_TRANSPORT_GRID_NX/NY/NZ`	Light-history voxel resolution; storage grows with their product.
`REPRESENTATIVE_RAYS / MAX_SEGMENTS_PER_RAY`	Trajectory storage limits; segment overflow truncates saved trajectories.
`SAVE_LIGHT_TRANSPORT_DATA / VISUALISATIONS`	Enable raw products and rendering; both False disables this collection path.

Resolution is not just DPI

PLOT_DPI controls raster export. The diagnostic voxel grid and slice thickness control spatial sampling. Display percentiles, log floor, cloud voxel cap, alpha and marker size control appearance only; they do not change exported raw arrays. The supplied light-history plots use greyscale, increasing brightness with density.

Memory planning

The four default 256 × 256 × 384 volume fields require about **384 MiB** before trajectory and other buffers (three float32 fields and one uint32 field). Production ray buffers, Mie tables, sediment fields, transfers and plotting need additional GPU/host memory. This is not a total-memory estimate.

Halving all three voxel dimensions reduces those volume fields by a factor of eight. Reducing only diagnostic ray count reduces observed work, not fixed grid storage. Reduce CHUNK_SIZE for transport buffers; reduce grid dimensions for volume buffers. The code rejects diagnostic requests beyond its allocation budget, but that is not a guarantee of available device memory.

Interpret maps cautiously

Track-length density is normalised per diagnostic launched ray and voxel volume; it has units m^-2 ray^-1. Event density has units m^-3 ray^-1. Segment totals cover the diagnostic grid, not full physical path length. A truncated representative trajectory is not a failed production ray.

USER GUIDE / 12

Find and read the output files

Directory convention

```
claritas_output/
  YYYYMMDD_HHMMSS/
    loess/
      622nm/
        0p5gL/
        2gL/
        4gL/
    kaolin/
      622nm/
        0p5gL/
```

Timestamped sessions are enabled by default. Decimals use p in folder names (0.5 becomes 0p5). If a case directory already exists, a _run02-style suffix prevents overwrite. The output base is resolved from the working directory, so run from the project folder or specify OUTPUT_DIR explicitly.

File / group	Read it for
`run_config.json`	Resolved geometry, sediment, optics, seed, model_kwargs, policy and source hash.
`run_summary.json / simulation_summary.csv`	Requested/launched rays, coefficients, detector totals, failure and diagnostic assessments.
`detector_response.csv`	Absolute hardware scores, normalized shape, native/mirror counts, approximate absolute error and jackknife normalized error.
`particle_optical_properties.csv`	Per-bin optical properties and coefficients.
`phase_function.csv`	Reporting Mie average CDF/PDF; not the hybrid production law.
`sediment_* / sensor_ring_* CSVs`	Stationary fields, size profiles and local optical environment; detailed-export rules apply.
`vortex_field.csv`	Prescribed free-surface/velocity profile and derived geometry.
`light_transport_volume.npz`	Raw volume arrays, coordinates and diagnostic normalizer; array order [z,y,x].
`light_transport_* CSVs and PNGs`	Sections, trajectories and display/integrity metadata.
`sweep_* / selected_angles_summary.csv`	Aggregated completed-case outputs when multiple cases finish.

SAVE_CSV controls many ordinary CSV exports. Light-history data use their own switches. JSON integrity/configuration reporting is not disabled by turning off ordinary CSV export. Missing detailed sediment plots in later sweep cases may be intentional.

USER GUIDE / 13

Integrity, uncertainty and result interpretation

Read these reports before using the numbers

Report	Meaning
`production_integrity_report.json`	Launched/failed counts, categories, samples, denominator, status and case-stop decision.
`water_boundary_report.json`	Water advance guard failures and available state samples.
`light_transport_integrity_report.json`	Field validity, event reconciliation, closure, limited outputs and policy decision.
`light_transport_summary.json`	Diagnostic normalisation, limits, memory, truncation and rendering metadata.
`case_failure_report.json / sweep_progress.json`	Fatal exception and completed-case checkpoint; not a resume file.

Failure policy

DIAGNOSTIC_POLICY="report" is the default. Isolated diagnostic faults can produce limited products while valid production reporting continues. In strict mode, integrity faults request a case stop after saved reports and safe outputs. MAX_FAILED_RAY_FRACTION=0.001 is enforced in either mode. ABORT_ON_SYSTEMIC_DIAGNOSTICS=False means diagnostic-only systemic flags do not automatically stop report-mode production.

Invalid configuration, corrupted array shapes/counts, nonfinite or negative production detector fields, and CUDA/programming errors remain fatal. Nonfinite or negative diagnostic arrays can suppress the standard derived bundle; trustworthy raw fields may be saved in light_transport_safe_raw_fields.npz. Values are not silently repaired.

Denominators and closure

Failed rays stay in launched-ray denominators. Valid prefixes may remain in diagnostic fields. Event reconciliation uses deposited + rejected - observed, with a separate check against the voxel sum. Boundary adjustment counts alone are informational. Policy thresholds are operational limits, not experimentally proven accuracy limits.

Detector uncertainty and scientific claims

Absolute detector export uses hardware symmetry scores; normalized_score divides by the summed hardware score. The absolute sqrtN column is an approximation, while normalized errors use batch jackknife estimates. A JSON null or empty uncertainty cell can mean an estimate is unavailable; it is not necessarily zero.

Normalized response addresses angular shape. Absolute amplitude requires an independent experimental basis. Comparison-match flags only compare settings. parabolic_transport_validation_pass is a limited failure/finite-majorant check and must not be presented as complete physical validation.

USER GUIDE / 14

Troubleshooting

Symptom	Next action
ModuleNotFoundError: cupy / numpy	Print sys.executable. Select and activate the intended environment. Install supplied requirements only if absent; do not install into an unrelated Python.
conda not recognised / activation fails	Open the installed Conda prompt. Initialise the intended shell, reopen the terminal, and select the matching VS Code interpreter.
No usable CUDA device / invalid GPU_DEVICE	Run nvidia-smi and the device-count probe. Check driver, environment and zero-based device index.
NVRTC compilation or missing CUDA headers	Inspect cp.show_config() and the full traceback. Check the CUDA-12 CuPy wheel/component installation and conflicting CuPy packages. A generic CUDA_PATH warning alone is not the diagnosis.
CUDA out of memory	Check nvidia-smi for other GPU use. Reduce CHUNK_SIZE and diagnostic grid/trajectory storage. Lowering ray count alone may not reduce the fixed diagnostic grid.
Stationary solver does not converge	Read residual/iteration error and saved settings. Review vortex/circulation, grid, iteration budget and tolerance. Do not relax a tolerance just to label a run successful.
CaseIntegrityError after outputs	Inspect production and diagnostic reports. Files can be saved before the policy requests a stop.
File not found / output in unexpected place	Check terminal working directory and extracted folder nesting. Use quoted paths. Output is relative to the working directory unless made explicit.
Plot backend / display error on Linux	Use SHOW_PLOTS=False; for a headless session export MPLBACKEND=Agg. Retain the traceback if an error persists.
Weak/no rear-angle score in short run	Inspect counts and failure reports, then assess sampling with larger runs and independent seeds. Do not infer an amplitude fit from a normalized sparse plot.

A useful support bundle

Keep the console traceback/log, run_config.json, run_summary.json, production/light integrity reports, water_boundary_report.json, environment package list and GPU/driver information. Include the actual source hash. An interrupted or fatal case may lack some products; say which are absent.

No automatic restart or successful-ray reweighting is provided. Preserve the original report before changing configuration to investigate a failure.

Configuration reference: how to use the appendix

The following tables list every assignment in the USER CONFIGURATION section as supplied, extracted directly from the cleaned code. They are defaults, not compulsory settings and not a record of a run. Editing values and starting a new Python process is the normal configuration workflow.

Controls that belong to different layers

Layer	Controls / implications
Production sampling	N_RAYS, seed, GPU_DEVICE, CHUNK_SIZE, statistics batch size.
Production physics	Geometry, density/viscosity/gravity, optics, vortex and stationary transport.
Production sediment discretisation	Quadrature, radial lookup, meridional grid and solver controls.
Diagnostic collection	Light-history ray limit, voxel dimensions, representative-ray and segment caps.
Reporting sampling	SEDIMENT_OUTPUT_NR/NZ, vertical/profile height points, profile radii and fractions.
Rendering only	DPI, display percentiles, cloud cap, marker size, alpha and surface margin.
Export and integrity	Save switches, output session policy and operational failure budgets.

Important constraints

• Keep the source, wall, ring and bore geometry consistent. Sensor height must remain below the mean water level; vortex centre must remain above the bottom and wall surface below wall top.

• Use positive eddy diffusivity, viscosity, gravity and water density. Particle density must exceed water density in the configured transport model.

• Steady meridional transport needs an enabled nonzero vortex. Circulation ratio lies in [0, 0.5].

• Diagnostic voxel dimensions must each be at least 4. Representative rays must not exceed the diagnostic ray limit, and segment capacity must be positive.

• Display percentiles obey 0 <= low < high <= 100; sizes, alpha and output-grid controls have their own configuration checks.

Constructor settings are a separate reference

Additional optical-event, source, surface, detector and aggregation options live in MODEL_DEFAULT_KWARGS and the model constructor, not in the top USER CONFIGURATION block. _build_model overrides matching physical settings from the top block. Change advanced options deliberately and verify run_config.json model_kwargs, rather than assuming a template edit survived an override.

USER GUIDE / 16

All user defaults / 1 of 5

Exact assignments from the delivered USER CONFIGURATION section. Long names wrap visually; copy the spelling from the source when editing.

Setting	Supplied value
DIAGNOSTIC_POLICY	'report'
MAX_FAILED_RAY_FRACTION	0.001
MAX_UNEXPLAINED_EVENT_FRACTION	0.001
MAX_DIAGNOSTIC_CLOSURE_ERROR	0.001
ABORT_ON_SYSTEMIC_DIAGNOSTICS	False
STRICT_DIAGNOSTIC_CLOSURE_ERROR	2e-06
RUN_SWEEP	True
MATERIAL	'loess'
CONCENTRATION_G_L	0.5
WAVELENGTH_NM	622.0
N_RAYS	100000000
RANDOM_SEED	24620001
GPU_DEVICE	0
MATERIALS	['loess', 'kaolin']
CONCENTRATIONS_G_L	[0.5, 2.0, 4.0]
WAVELENGTHS_NM	[622.0]
USE_SAME_SEED_FOR_SWEEP	False
SEDIMENT_DISTRIBUTION	'steady_state'
SEDIMENT_TRANSPORT_MODEL	'schiller_naumann_meridional_advection_diffusion'
SEDIMENT_EDDY_DIFFUSIVITY_M2_S	0.001

N_RAYS-dependent defaults are expressions evaluated when the module loads. None disables selected policy limits or selects the geometry/optical behavior explained in the main guide; it is not interchangeable across controls.

USER GUIDE / 17

All user defaults / 2 of 5

Exact assignments from the delivered USER CONFIGURATION section. Long names wrap visually; copy the spelling from the source when editing.

Setting	Supplied value
SEDIMENT_PARTICLE_DENSITY_KG_M3	2600.0
WATER_DENSITY_KG_M3	997.0
WATER_DYNAMIC_VISCOSITY_PA_S	0.00089
GRAVITY_M_S2	9.81
MERIDIONAL_CIRCULATION_RATIO	0.5
SEDIMENT_TRANSPORT_QUADRATURE_POINTS	4096
SEDIMENT_RADIAL_LOOKUP_POINTS	2049
MERIDIONAL_GRID_NR	48
MERIDIONAL_GRID_NZ	96
MERIDIONAL_SOLVER_MAX_ITERATIONS	8000
MERIDIONAL_SOLVER_CHECK_INTERVAL	250
MERIDIONAL_SOLVER_TOLERANCE	1e-06
MERIDIONAL_SOLVER_SOR	1.75
ENABLE_VORTEX	True
VORTEX_MODEL	'localized_scully_vortex'
VORTEX_WALL_CENTER_HEIGHT_DIFFERENCE_M	0.05
VORTEX_CORE_RADIUS_M	0.02
SAMPLE_CELL_DIAMETER_M	0.098
SAMPLE_CELL_HEIGHT_M	0.491
WATER_LEVEL_M	0.142

N_RAYS-dependent defaults are expressions evaluated when the module loads. None disables selected policy limits or selects the geometry/optical behavior explained in the main guide; it is not interchangeable across controls.

USER GUIDE / 18

All user defaults / 3 of 5

Exact assignments from the delivered USER CONFIGURATION section. Long names wrap visually; copy the spelling from the source when editing.

Setting	Supplied value
SENSOR_RING_HEIGHT_M	0.093
ACRYLIC_WALL_THICKNESS_M	0.0035
SENSOR_RING_INNER_CLEARANCE_M	0.0005
SENSOR_RING_RADIAL_WIDTH_M	0.015
COUNTERBORE_DEPTH_M	0.008
THROUGH_BORE_DIAMETER_M	0.004
COUNTERBORE_DIAMETER_M	0.0087
BOTTOM_DISC_THICKNESS_M	0.003
SOURCE_LAUNCH_RADIUS_M	None
WATER_REFRACTIVE_INDEX	1.33
ACRYLIC_REFRACTIVE_INDEX	1.4906
AIR_REFRACTIVE_INDEX	1.0
PARTICLE_N_REAL_OVERRIDE	None
PARTICLE_K_IMAG_OVERRIDE	0.0
PARTICLE_SURFACE_RMS_SLOPE_DEG	25.0
WAVE_SURFACE_RMS_HEIGHT_NM	100.0
OUTPUT_DIR	'claritas_output'
CREATE_TIMESTAMPED_SESSION	True
SAVE_CSV	True
SAVE_PLOTS	True

N_RAYS-dependent defaults are expressions evaluated when the module loads. None disables selected policy limits or selects the geometry/optical behavior explained in the main guide; it is not interchangeable across controls.

USER GUIDE / 19

All user defaults / 4 of 5

Exact assignments from the delivered USER CONFIGURATION section. Long names wrap visually; copy the spelling from the source when editing.

Setting	Supplied value
SHOW_PLOTS	False
SAVE_PHASE_FUNCTION_CSV	True
EXPORT_SEDIMENT_FIELDS	True
EXPORT_LOCAL_PSD_PROFILES	True
SAVE_SEDIMENT_DIAGNOSTIC_PLOTS	True
SAVE_PSD_DIAGNOSTIC_PLOTS	True
DETAILED_DIAGNOSTICS_PER_SWEEP_CASE	False
SAVE_VORTEX_PROFILE	True
SUMMARY_ANGLES_DEG	[0, 20, 90, 120, 140, 170]
SAVE_LIGHT_TRANSPORT_VISUALISATIONS	True
SAVE_LIGHT_TRANSPORT_DATA	True
LIGHT_TRANSPORT_DIAGNOSTIC_MAX_RAYS	N_RAYS
LIGHT_TRANSPORT_GRID_NX	256
LIGHT_TRANSPORT_GRID_NY	256
LIGHT_TRANSPORT_GRID_NZ	384
LIGHT_TRANSPORT_REPRESENTATIVE_RAYS	256
LIGHT_TRANSPORT_MAX_SEGMENTS_PER_RAY	512
LIGHT_TRANSPORT_VERTICAL_SLICE_THICKNESS_M	0.001
LIGHT_TRANSPORT_SENSOR_SLICE_THICKNESS_M	0.001
LIGHT_TRANSPORT_LOG_FLOOR	1e-12

N_RAYS-dependent defaults are expressions evaluated when the module loads. None disables selected policy limits or selects the geometry/optical behavior explained in the main guide; it is not interchangeable across controls.

USER GUIDE / 20

All user defaults / 5 of 5

Exact assignments from the delivered USER CONFIGURATION section. Long names wrap visually; copy the spelling from the source when editing.

Setting	Supplied value
LIGHT_TRANSPORT_PLOT_DPI	400
LIGHT_TRANSPORT_DISPLAY_LOW_PERCENTILE	2.0
LIGHT_TRANSPORT_DISPLAY_HIGH_PERCENTILE	99.5
LIGHT_TRANSPORT_3D_MIN_DISPLAY_PERCENTILE	5.0
LIGHT_TRANSPORT_3D_MAX_DISPLAY_VOXELS	2000000
LIGHT_TRANSPORT_3D_ALPHA_MIN	0.06
LIGHT_TRANSPORT_3D_ALPHA_MAX	0.5
LIGHT_TRANSPORT_3D_MARKER_SIZE_MIN	0.15
LIGHT_TRANSPORT_3D_MARKER_SIZE_MAX	0.8
LIGHT_TRANSPORT_3D_SURFACE_MARGIN_M	0.005
SEDIMENT_PROFILE_RADII_FRACTION	[0.0, 0.5, 0.9]
SEDIMENT_PROFILE_HEIGHT_POINTS	200
PSD_PROFILE_HEIGHT_FRACTIONS	[0.05, 0.25, 0.5, 0.75, 0.95]
SEDIMENT_OUTPUT_NR	64
SEDIMENT_OUTPUT_NZ	96
VORTEX_OUTPUT_NR	256
CHUNK_SIZE	250000
STATISTICS_BATCH_RAYS	250000

N_RAYS-dependent defaults are expressions evaluated when the module loads. None disables selected policy limits or selects the geometry/optical behavior explained in the main guide; it is not interchangeable across controls.

USER GUIDE / 21

Sources and verification scope

Authoritative implementation

The guide is based on the complete cleaned CLARITAS.py, its supplied requirements.txt, README.md, CLEANUP_REPORT.md, validate_claritas.py and test_diagnostics.py. Code is authoritative if this guide is used with a modified file.

CLARITAS.py SHA-256:

```
95d056e6de2089be6290bd1fbe99e1c49c8c94f8e1590353cef788e5be96da33
```

What was checked

The cleaned source passed host/static tests, diagnostic-policy tests, numerical equivalence checks and CPU-compiled boundary-helper checks during cleanup. The current bundled validator has 16 host/static checks and four optional GPU checks; the policy suite has 10 tests. These checks do not establish absolute optical amplitude accuracy.

Guide configuration examples were checked against the actual configuration resolver. Python snippets were syntax-checked. The PDF was rendered for page-by-page layout inspection. Windows GUI/setup instructions and CUDA execution were not exercised in this Linux documentation environment; external setup instructions were checked against official documentation.

Official setup references

[1] CuPy installation and CUDA component wheels

https://docs.cupy.dev/en/stable/install.html

[2] VS Code Python environments and interpreter selection

https://code.visualstudio.com/docs/python/environments

[3] VS Code Windows installation

https://code.visualstudio.com/docs/setup/windows

[4] Miniforge installers and documentation

https://github.com/conda-forge/miniforge

[5] Conda shell initialisation

https://docs.conda.io/projects/conda/en/stable/commands/init.html

[6] VS Code Linux installation

https://code.visualstudio.com/docs/setup/linux

[7] NVIDIA Linux driver/toolkit installation guidance

https://docs.nvidia.com/cuda/cuda-installation-guide-linux/index.html

[8] NVIDIA Windows installation guidance

https://docs.nvidia.com/cuda/cuda-installation-guide-microsoft-windows/index.html

Setup references checked 11 October 2026. The supplied dependency bounds remain unchanged. Use driver guidance appropriate to your GPU and operating system; the Python-wheel setup does not require the full system toolkit route.